SPL 2

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An urban Issue Mapping Platform



Team 02

Team Members

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Problems

- **No Centralised** platform for issue reporting
- **No Geospatial** tracking



- **Serious issues** gets burried
- **No organised way** to helpout as citizens

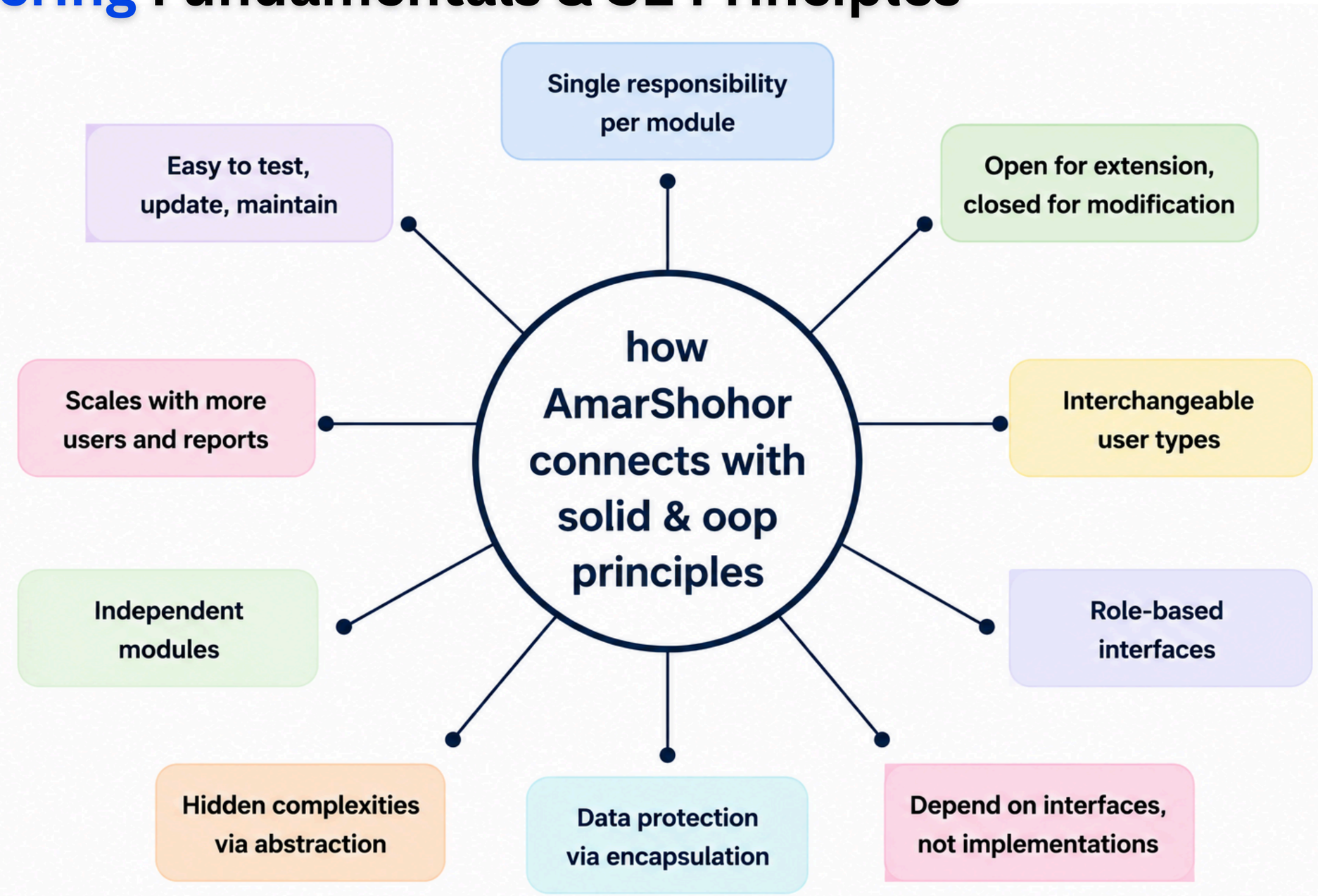
Stakeholders:

- **Primary:** Citizens, Volunteers
- **External:** Emergency services(police, fire, ambulance), Notification gateways, Map provider, General public

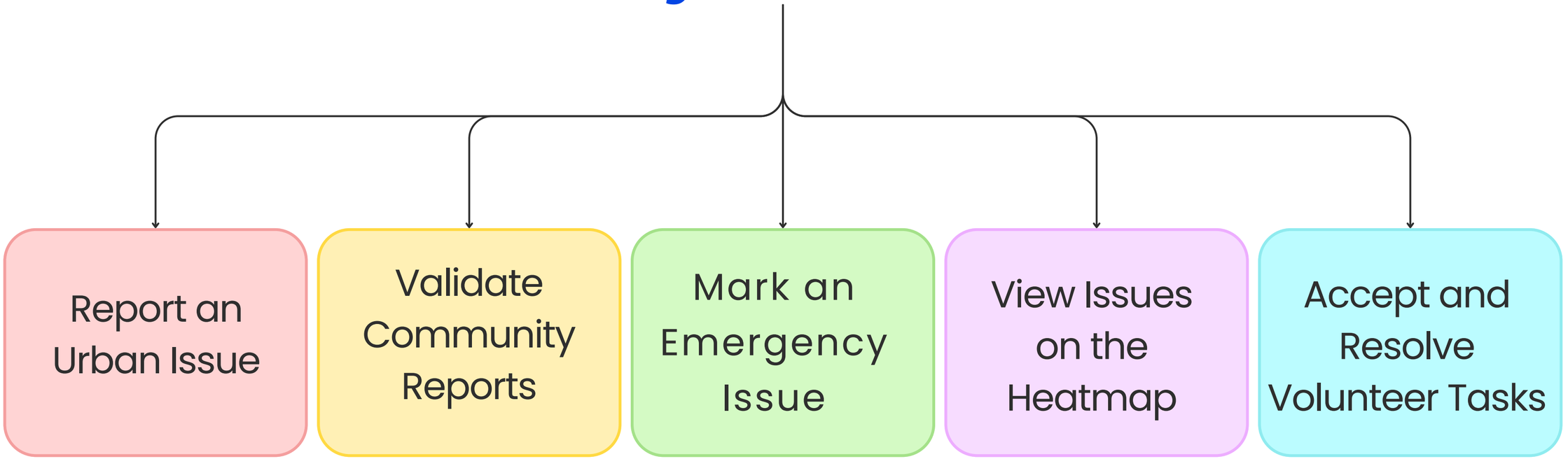
Requirement Gathering Methods:

- Interviews
- Questionnaires
- Analysis of similar platforms

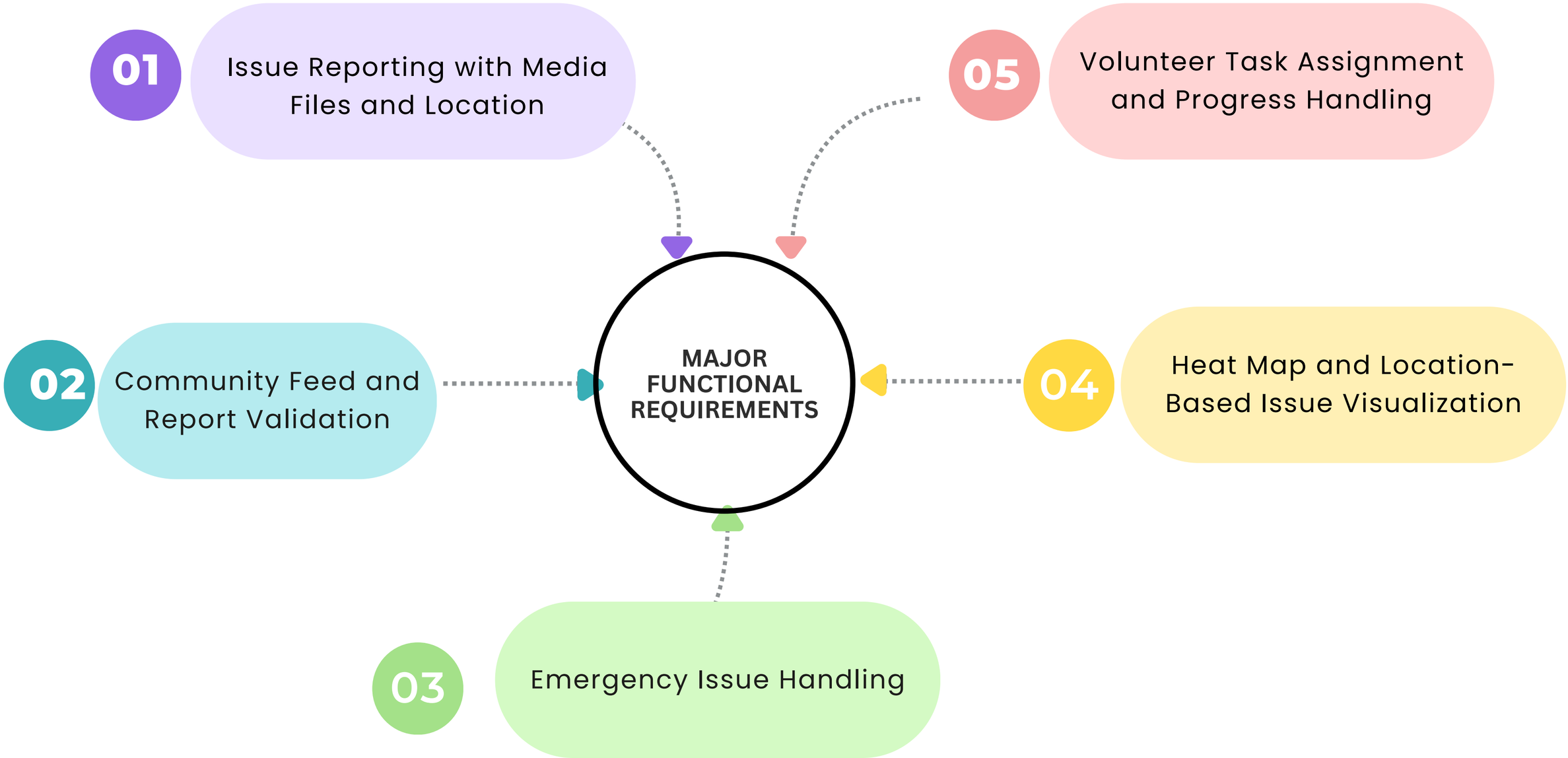
Engineering Fundamentals & SE Principles



Summary of User Stories



Summary of Major Functional Requirements


01
Issue Reporting with Media Files and Location
05
Volunteer Task Assignment and Progress Handling
MAJOR FUNCTIONAL REQUIREMENTS
02
Community Feed and Report Validation
04
Heat Map and Location-Based Issue Visualization
03
Emergency Issue Handling

Module 1
Issue Reporting

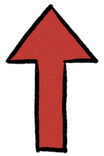
- Select **Issue Category**
- add **specific location**
- Upload **photos** & **videos** as evidence
- Prevent **Duplicate** reporting of issues

Track **Issue cycle**:



Module 2
Community Feed & Validation System

- Reported issues appear on Community Feed



- Upvote or Report as spam



- Comment on Community Reports

Module 3

Emergency Issue Handling

Emergency Incident Reporting

Emergency reports receive high priority

Nearby volunteers are notified immediately

Fire **Outbreak**





Road **blockade**

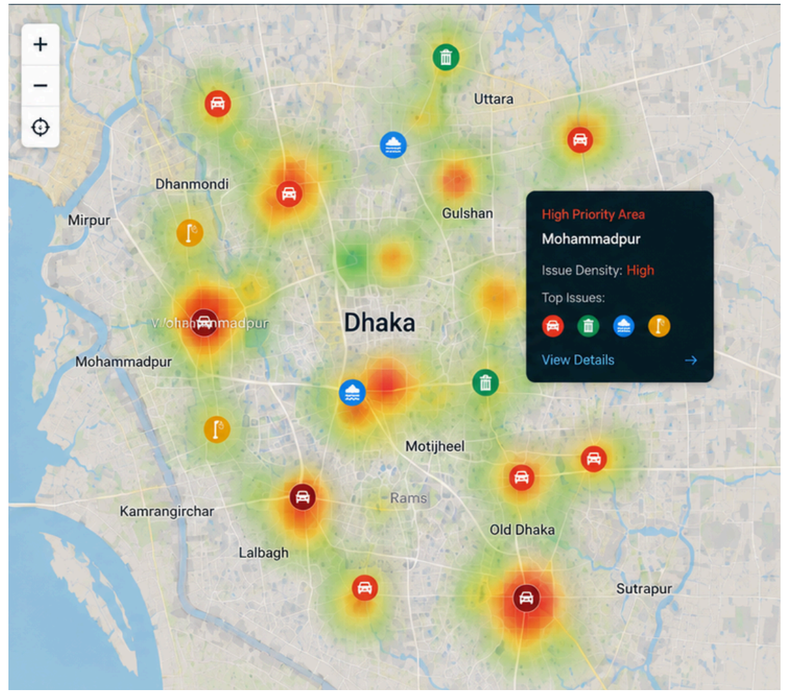
Module 4

HeatMap Tracking

Generate heatmap from reported locations

Update the heatmap as new reports are submitted

Users click hotspots to view associated reports



Module 5
Volunteer Task Assigning

Match volunteers based on Skills, Location, Availability

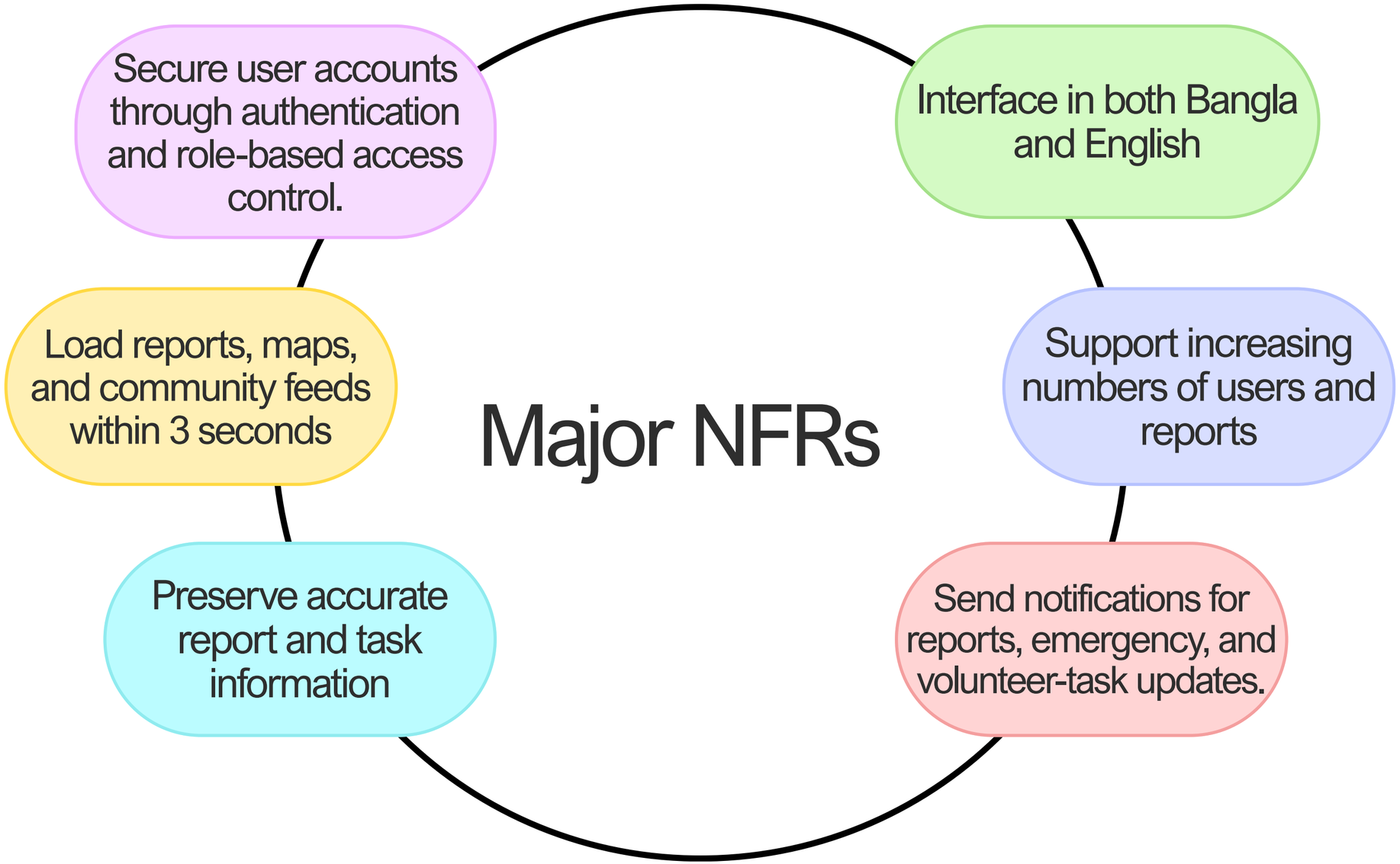
Volunteers request or accept tasks

Earn Achievement Badges for Successfully Completed Tasks

Notify citizens when task status changes






Secure user accounts through authentication and role-based access control.
Interface in both Bangla and English
Load reports, maps, and community feeds within 3 seconds
Support increasing numbers of users and reports
Major NFRs
Preserve accurate report and task information
Send notifications for reports, emergency, and volunteer-task updates.

THANK YOU

EDGE CASES


DUPLICATE REPORTS
SCAM/FAKE REPORTS
Incompletition of Volunteer tasks
Multiple Volunteers assignment
False Evidence Upload
Low Population Areas
Location Errors
Citizens don’t confirm resolution